

Standard Authorization Request (SAR)

Complete and submit this form, with attachment(s) to the [NERC Help Desk](#). Upon entering the Captcha, please type in your contact information, and attach the SAR to your ticket. Once submitted, you will receive a confirmation number which you can use to track your request.

The North American Electric Reliability Corporation (NERC) welcomes suggestions to improve the reliability of the bulk power system through improved Reliability Standards.

Requested information			
SAR Title:		Reliability Standards to Address Computational Load Disturbance Performance	
Date Submitted:			
SAR Requester			
Name:	Eric Meier, ERCOT (NERC LLWG Chair) Sudip Manandhar, Southern Company (NERC LLWG Vice Chair) Rahul Anilkumar (Danovo Energy Solutions) Evan Mickelson, NERC (NERC LLWG Secretary) Jack Gibfried, NERC (NERC LLWG Secretary) Submitted on behalf of the NERC Large Loads Working Group (LLWG)		
Organization:	NERC Large Loads Working Group (LLWG)		
Telephone:	Eric Meier: 512-248-6770 Sudip Manandhar: 404-416-3176 Rahul Anilkumar: 510-309-8654 Evan Mickelson: 770-843-8815 Jack Gibfried: 470-432-8484	Email:	eric.meier@ercot.com smanandh@southernco.com RAnilkumar@danovoenergy.com Evan.Mickelson@nerc.net Jack.Gibfried@nerc.net
SAR Type (Check as many as apply)			
☒ New Standard ☒ Revision to Existing Standard ☒ Add, Modify or Retire a Glossary Term ☐ Withdraw/retire an Existing Standard		☐ Imminent Action/ Confidential Issue (SPM Section 10) ☐ Variance development or revision ☐ Other (Please specify)	
Justification for this proposed standard development project (Check all that apply to help NERC prioritize development)			
☐ Regulatory Initiation ☒ Emerging Risk (Reliability Issues Steering Committee) Identified ☐ Reliability Standard Development Plan		☒ NERC Standing Committee Identified ☐ Enhanced Periodic Review Initiated ☒ Industry Stakeholder Identified	
What is the risk to the Bulk Electric System (What Bulk Electric System (BES) reliability benefit does the proposed project provide?):			
This project will help address the risks posed to the Bulk Power System (BPS) by the disturbance response of computational loads, which have reduced or interrupted demand in response to transmission faults.			

Requested information

NERC's Reliability and Security Technical Committee (RSTC) established the Large Load Task Force (LLTF), now the Large Loads Working Group (LLWG), to analyze the reliability impacts of emerging large loads. The LLWG identified the lack of predictable ride-through performance for large loads as a gap in the current NERC Reliability Standards in its [Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads](#) white paper (March 2026). NERC issued a [Level 2 Alert](#) in September 2025 addressing rapid, major load swings and suggesting changes to interconnection requirements, modeling practices, and operation readiness. NERC also issued a [Level 3 Alert](#) in May 2026 containing essential actions for the reliable integration of computational loads. [Project 2026-02 Computational Loads](#) was initiated in March 2026 to address in the near term the most critical issues associated with integrating computational loads.

Project 2026-02 scope addresses computational load interconnection and modeling, operational coordination and information, protection and control systems, and event analysis information. Ride-through and post-fault active power recovery (PFAPR) performance requirements are technically distinct from that near-term scope and rest on the event analysis and dynamic simulation basis set out in the preliminary LLWG and Event Analysis Subcommittee (EAS) white paper [Large Load Disturbance Performance - Impact Analysis and Ride-Through Recommendations](#). The Project 2026-02 project is also facing a regulatory deadline of December 31, 2026 for filing with the Federal Energy Regulatory Commission. This SAR is submitted so that the needs associated with disturbance performance, as the next most high-priority issue associated with computational load integration, receive a dedicated drafting effort.

The following event records establish the BPS impact of these loads and the technical justification for a SAR on this matter.

- On July 10, 2024, the failure of a lightning arrestor on a 230 kV line in the Dominion Energy Virginia (DEV) service territory produced a single-line-to-ground fault, that ultimately led to a line lockout after a staggered 3-attempt automatic reclosing sequence at each terminal (i.e., 6 fault applications within approximately 82 seconds). While, no data center was disconnected by the fault itself or by the reclosing sequence, approximately 1,500 MW of data center load nevertheless transferred to backup generation through customer-side controls and protection and remained off the BPS for several hours, producing a measurable frequency deviation, a change in area control error, and a localized overvoltage condition that persisted at several substations until DEV system operators energized shunt reactors and removed capacitor banks from service. This load transfer is considered to be customer-initiated load reduction (CILR).¹ A subsequent event in the same area on February 17, 2025 produced approximately 1,800 MW of CILR across 18 substations and 44 transformers. Recently on July 22, 2026, an event resulted in approximately 3,800 MW of CILR across Northern Virginia, and more than double the CILR observed during the July 10, 2024, event.²

¹ See [Possible Misunderstandings of the Term "Load Loss"](#)

² <https://www.pjm.com/-/media/DotCom/committees-groups/subcommittees/sos/2026/20260730/20260730-item-06---july-22-dominion-load-transfer-event---presentation.pdf>

Requested information

- In the Texas Interconnection, ERCOT has analyzed 32 large load ride-through events with CILR of 100 MW or greater between January 2023 and April 2026 and has indicated that the simultaneous loss of 3,200 GW of load under certain system conditions may pose significant risk to Interconnection frequency and voltage stability and may require ERCOT to lower system operating limits on transmission interfaces in affected areas.
- AEP has experienced two significant data center CILR events. In February 2025, a 428 MW CILR event occurred. In March 2025, a 227 MW CILR event occurred. More information on the ERCOT events, the AEP events, and the DEV events can be found in the preliminary white paper [Large Load Disturbance Performance – Impact Analysis and Ride-Through Recommendations](#).

Moreover, the observed response is not consistent between computational load sites or between events at the same site. Sites connected to the same substation and exposed to nearly identical voltage depressions have behaved differently. For example, a single site reduced approximately 20% of its load at a minimum phase voltage of 0.7205 pu during one event and its full load at a shallower minimum phase voltage of 0.7538 pu during another. Planning Coordinators (PC), Transmission Planners (TP), and Balancing Authorities (BA) therefore cannot presently predict how much load a credible fault will remove from service.

[PRC-024-4](#) establishes frequency and voltage protection settings for synchronous generators, Type 1 and Type 2 wind resources, and synchronous condensers, and [PRC-029-1](#) establishes ride-through requirements for inverter-based resources (IBR). Neither applies to load, and no NERC Reliability Standard establishes a minimum voltage ride-through (VRT), frequency ride-through (FRT), or PFAPR performance requirement for computational load.

Purpose or Goal (What are the reliability gap(s) or risk(s) to the Bulk Electric System being addressed, and how does this proposed project provide the reliability-related benefit described above?):

The goal of this project is to address the reliability risk posed by the computational loads performing CILR by defining minimum ride-through, rate-of-change-of-frequency (RoCoF) withstand, and PFAPR performance requirements to ensure the stability of the BPS. When the evidentiary record of BPS CILR disturbances is considered in light of the growing queues of computational loads seeking interconnection, it highlights that it is essential to address issues associated with disturbance performance to support integration of computational loads in a way that avoids placing the reliability of the BPS at risk.

To mitigate the risks, two actions are needed: (1) revisions to the *NERC Glossary of Terms Used in NERC Reliability Standards* (NERC Glossary of Terms) to define load ride-through and PFAPR, consistent with the computational load and Computational Load Entity definitions under development in Project 2026-02; and (2) the development of one or more Reliability Standards establishing minimum voltage ride-through (VRT), frequency ride-through (FRT), RoCoF withstand, and PFAPR performance requirements for computational loads.

Establishing these disturbance performance requirements will ensure that applicable computational loads have clear performance requirements to meet or exceed, while preserving the ability of the Transmission Owner, Distribution Provider, and Generator Owner, in coordination with other applicable entities as needed, to establish more stringent performance requirements through their interconnection

Requested information

requirements. Minimum disturbance performance requirements will also give equipment manufacturers and computational load site developers a clear, consistent and upfront design target in place of the regional and utility-specific requirements now in development. The clear and consistent design targets reduce configuration risk.

Project Scope (Define the parameters of the proposed project):

The scope of this project shall include:

- Develop new or modified Reliability Standards addressing computational load ride-through, PFAPR, RoCoF withstand, and current consumption during disturbances.
- Developing terms and definitions for inclusion in the *NERC Glossary of Terms* that support Reliability Standards addressing computational load ride-through and PFAPR.
- Considering the LLWG and EAS white paper *Large Load Disturbance Performance: Impact Analysis and Ride-Through Recommendations* (preliminary version available on the [NERC website](#)); the [Level 3 Alert](#) Essential Actions; the requirements published or proposed by ERCOT, SPP, MISO, Southern Company, DEV, AESO, and Fingrid; and other information and stakeholder feedback; develop one or more Reliability Standards establishing minimum VRT, FRT, RoCoF withstand, PFAPR, and disturbance current consumption requirements for computational loads.
- Determining applicability, including the point of measurement, the treatment of cooling load, the performance verification and disturbance monitoring data required to demonstrate compliance, and a tiered applicability and effective date structure addressing sites that are already operational or that hold executed interconnection agreements.
- Addressing the coordination of computational load ride-through obligations with the generation ride-through obligations applicable to co-located generation sharing a point of common coupling (PCC). Determining if single net PCC performance region should replace separate load and generation ride-through obligations at sites where load and generation are tightly integrated.

Detailed Description (Describe the proposed deliverable(s) with sufficient detail for a drafting team to execute the project. If you propose a new or substantially revised Reliability Standard or definition, provide: (1) a technical justification³ of developing a new or revised Reliability Standard or definition, which includes a discussion of the risk and impact to reliability-of the BES, and (2) a technical foundation document (e.g., research paper) to guide development of the Standard or definition):

This drafting team will develop definitions for inclusion in the *NERC Glossary of Terms* supporting the requirements described below, including a term for load ride-through and a term for PFAPR. The drafting team should consider whether the development of a new load-specific ride-through term is necessary rather than modification of the existing Ride-through definition approved with [PRC-024-4](#) and [PRC-029-1](#). In addition, the drafting team should modify or develop one or more Reliability Standards to address the following disturbance performance and mitigation requirements.

³ The NERC Rules of Procedure require a technical justification for new or substantially revised Reliability Standards. Please attach pertinent information to this form before submittal to NERC.

Requested information

- Voltage Ride-Through:** Establish minimum VRT performance requirements applicable to computational loads, including requirements for low-voltage and high-voltage conditions. Requirements should define the voltage-time performance envelope (e.g., expressed as a table of voltage magnitudes and corresponding minimum ride-through durations) applicable to computational loads to support BPS reliability during system disturbances. The LLWG recommends consideration of the materials above, and recommends drafting team consideration of the ERCOT Voltage Ride-Through Requirements for Large Computational Loads from [ERCOT NOGRR 282](#) as the minimum baseline, modified to extend the obligation across the full duration of credible events and fault clearing times, and modified to reduce the number of discrete voltage-time steps. Table 1 provides a recommended starting point for these modifications. The drafting team should also address the applicability of VRT requirements to cooling load providing some flexibility where a site demonstrates through study that its cooling configuration cannot practically meet the requirements and the residual risk is acceptable for BPS reliability.

Voltage (per unit)	Operation Region*	Minimum Ride-Through Time (sec)
≥ 1.15	May Ride Through or Trip	N/A
< 1.15 and ≥ 0.90	Continuous operation region	Continuous
$0.80 < V \leq 0.90$	Mandatory operation region	2.00
$0.50 \leq V \leq 0.80$	Mandatory operation region	0.50
$V < 0.50$	Mandatory operation region	0.25

*Mandatory operation region would need to clarify if the computational load would need to stay connected, but may reduce active power proportional to voltage drop.

- Multi-Disturbance Ride-Through:** Establish requirements that address the survivability of successive voltage depressions rather than a single fault application. Where a site uses protection logic that counts voltage dips within a predetermined window, the counter threshold should accommodate the automatic reclosing sequence expected on the transmission system serving the site, provided each individual event falls inside the applicable ride-through region.
- Frequency Ride-Through and RoCoF Withstand:** Establish minimum FRT and RoCoF withstand performance requirements at the Interconnection level, considering the [IEEE 2800-2022](#) and [PRC-029-1](#) criteria. The drafting team should consider whether the duration associated with the high-frequency requirement requires adjustment so that generation and load do not disconnect simultaneously during an over-frequency excursion. The drafting team should also consider existing under-frequency load shedding (UFLS) schemes to ensure that FRT requirements do not interfere with UFLS schemes.
- Post-Fault Active Power Recovery:** Establish requirements related to the restoration of load following a system disturbance, including restoration magnitude and recovery timing. The LLWG

Requested information

recommends a default recovery level of at least 90% of pre-disturbance consumption with entities being able to modify the recovery level and time based on system studies considering issues such as system strength, stability studies, and fault-induced delayed voltage recovery and other local area load dynamics. The LLWG also recommends that the requirement only applies to a site whose PCC voltage remained inside the ride-through region, and not a site whose PCC voltage fell outside that region and that consequently disconnected.

- **Current Consumption:** Establish requirements for the maximum current consumption of the computational load site during a disturbance.
- **Applicability and Effective Dates:** Develop criteria for applicability and implementation timelines, including consideration of both new and existing computational load sites. For comparison, [ERCOT NOGRR 282](#) provides a tiered model, in that it establishes a grandfathering date associated with operational status or execution of an interconnection agreement and then applies a mandatory compliance date to sites satisfying interconnection requirements after a later date. Computational load sites built before the grandfathering date must conform to the performance requirements unless they provide documentation of the technical inability to comply with said performance requirements. Every documented CILR event to date involved sites that were already operational, and a framework applicable only to sites entering the interconnection queue after the effective date would leave the identified risk unaddressed for the service life of the existing fleet.

Recommended Approach

The LLWG recommends the drafting team pursue the development of new or modified Reliability Standard(s) (with associated Glossary updates) addressing the disturbance performance requirements for computational loads, in the manner of [PRC-029-1](#) for IBRs.

The drafting team is not bound to the LLWG's recommended approach and may revise it or develop an alternative approach as informed by its expertise, technical information or analysis, or stakeholder feedback.

Technical Foundation and Justification

The technical foundation document for this project is the LLWG and EAS white paper *Large Load Disturbance Performance: Impact Analysis and Ride-Through Recommendations*. The preliminary version of this document is posted on the [NERC website](#). This white paper is planned to be finalized and published in December 2026 and is intended to help inform next steps (including any drafting team efforts). This document provides the event analysis, the review of published and proposed requirements across North American and international jurisdictions, and the recommendations on which the deliverables above are based.

Additional technical justification can be found in the LLWG [Characteristics and Risks of Emerging Large Loads](#) and [Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads](#) white papers, as well as the LLWG [Computational Loads Standard Authorization Request Roadmap](#).

Requested information
Cost Impact Assessment, if known (Provide a paragraph describing the potential cost impacts associated with the proposed project):
Exact costs for this project are unknown and are expected to vary by entity, by region, and by site design and vintage. Near-term costs are likely to be modest for sites designed after the requirements are known, as ride-through capability is largely a function of uninterruptible power supply (UPS) configuration, protection and control settings, disturbance counter logic, and transformer winding configuration, all of which are established during site engineering rather than through retrofit. Costs are expected to be higher for sites already in operation and for cooling infrastructure, which is typically not supported by the UPS and may require extension of that capability to pumps, fans, and chillers.
Please describe any unique characteristics of the BES facilities that may be impacted by this proposed standard development project (e.g., Dispersed Generation Resources):
This project proposes establishing performance requirements applicable to computational loads, and coordination obligations for entities who interface with such users. Other equipment at the computational load site may also be affected.
To assist the NERC Standards Committee in appointing a drafting team with the appropriate members, please indicate to which Functional Entities the proposed standard(s) should apply (e.g., Transmission Operator, Reliability Coordinator, etc. See the NERC Rules of Procedure Appendix 5A:
The requirements are expected to apply to the registered entity class under development for computational loads, with coordination obligations for the TO, TOP, TP, PC, BA, and RC. Drafting team members should have experience in computational load site electrical design with knowledge of facility characteristics that affect its disturbance performance, transmission protection and control (e.g., automatic reclosing practice, relay operating times), dynamic simulation and load modeling of power electronic load, and BA operations. Members should also include equipment manufacturer and computational load developer/owner representation so that the achievability of proposed requirements is represented during drafting.
Do you know of any consensus building activities ⁴ in connection with this SAR? If so, please provide any recommendations or findings resulting from the consensus building activity.
This SAR was developed by the NERC LLWG. The LLWG has issued two white papers and a Reliability Guideline analyzing the reliability impacts of emerging large loads, and NERC Event Analysis has published two incident reviews addressing voltage-sensitive load reductions in the DEV service territory and in the Texas Interconnection. The need for the SAR has also been widely discussed in Project 2026-02 drafting team meetings and associated outreach efforts. The preliminary white paper Large Load Disturbance Performance - Impact Analysis and Ride-Through Recommendations reflects consensus building already conducted and its recommendations are reflected in the deliverables described above.
Are there any related standards or SARs that should be assessed for impact as a result of this proposed project? If so, which standard(s) or project number(s)?

⁴ Consensus building activities are occasionally conducted by NERC and/or project review teams. They typically are conducted to obtain industry inputs prior to proposing any standard development project to revise, or develop a standard or definition.

Requested information

Yes. Identification of a Reliability Standard below does not propose its revision; the purpose is to confirm that the requirements developed under this SAR coordinate with the Reliability Standards already in place.

- PRC-024-4, as the colocation of computational load and generation could impact generator ride-through requirements if the drafting team determines that a single net PCC performance region should replace separate load and generation ride-through obligations.
- PRC-029-1, as the colocation of computational load and generation could impact generator ride-through requirements if the drafting team determines that a single net PCC performance region should replace separate load and generation ride-through obligations.

Are there alternatives (e.g., guidelines, white paper, alerts, etc.) that have been considered or could meet the objectives? If so, please list the alternatives with the benefits of using them.

The LLWG has published two white papers and a [Reliability Guideline](#) addressing emerging large loads, NERC Event Analysis has published two [incident reviews](#), and NERC issued a [Level 2 Alert](#) in September 2025 and a [Level 3 Alert](#) in May 2026. Requirements published or proposed outside NERC are already producing design change, including [ERCOT NOGRR 282](#), the [SPP High Impact Large Load framework](#), the [MISO large load proposal](#), and the [Southern Company](#) and [DEV interconnection requirements](#). The ERCOT requirements in particular have given equipment manufacturers a target to design toward. [IEEE 2800-2022](#) provides the FRT and RoCoF withstand basis adopted here but addresses IBRs rather than load. Grid-side mitigations (e.g., dynamic reactive support, faster protection, modified reclosing practice) reduce the severity of the disturbances that trigger CILR and remain necessary for residual risk.

Collectively these measures represent an important start, but they are voluntary or partial in coverage, the envelopes and recovery windows differ between regions, and none establishes an enduring performance requirement applicable to sites interconnecting in future years. Full mitigation requires a Reliability Standard establishing minimum ride-through and PFAPR performance.

Reliability Principles

Does this proposed standard development project support at least one of the following [Reliability Principles](#)? Please check all those that apply.

☒	1. Interconnected bulk power systems shall be planned and operated in a coordinated manner to perform reliably under normal and abnormal conditions as defined in the NERC Standards.
☒	2. The frequency and voltage of interconnected bulk power systems shall be controlled within defined limits through the balancing of real and reactive power supply and demand.
☒	3. Information necessary for the planning and operation of interconnected bulk power systems shall be made available to those entities responsible for planning and operating the systems reliably.
☐	4. Plans for emergency operation and system restoration of interconnected bulk power systems shall be developed, coordinated, maintained and implemented.
☐	5. Facilities for communication, monitoring and control shall be provided, used and maintained for the reliability of interconnected bulk power systems.
☐	6. Personnel responsible for planning and operating interconnected bulk power systems shall be trained, qualified, and have the responsibility and authority to implement actions.

Reliability Principles	
☐	7. The security of the interconnected bulk power systems shall be assessed, monitored and maintained on a wide area basis.
☐	8. Bulk power systems shall be protected from malicious physical or cyber attacks.

Market Interface Principles	
Does the proposed standard development project comply with all of the following Market Interface Principles ?	Enter (yes/no)
1. A reliability standard shall not give any market participant an unfair competitive advantage.	y
2. A reliability standard shall neither mandate nor prohibit any specific market structure.	y
3. A reliability standard shall not preclude market solutions to achieving compliance with that standard.	y
4. A reliability standard shall not require the public disclosure of commercially sensitive information. All market participants shall have equal opportunity to access commercially non-sensitive information that is required for compliance with reliability standards.	y

Identified Existing or Potential Regional or Interconnection Variances	
Region(s)/ Interconnection	Explanation
	None

For Use by NERC Only

Risk Tracking	
☐ Grid Transformation ☐ Resilience to Extreme Events ☐ Security	☐ Energy Policy ☐ Critical Infrastructure Interdependencies
CMEP Feedback	
☐ Yes	☐ No